Bhaskaracharya College of Applied Science

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Fundamental of Genetics report file

Topic: Eugenics

Can selective reproduction ever be scientifically or ethically justified?

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a. Introduction

Concept, scientific background, and relevance

What is eugenics?

An attempt to change the inherited features of a population by influencing who reproduces.

It has often involved ideas about “desirable” and “undesirable” people.

Scientific background

Traits can be inherited through genes. Francis Galton introduced the term eugenics.

But many traits also depend on environment and cannot be predicted by one gene.

Why is it relevant?

Genetic testing can help families understand inherited disease risks.

It also raises questions about choice, disability, fairness, and pressure on parents.

Key question: When does reducing disease risk become control over reproduction?

a. A short history

Why the past matters

1. **Galton’s ideas**

   He proposed encouraging the reproduction of people considered to have desirable inherited traits.

2. **Forced sterilization**

   Laws in the United States and other countries stopped some people from having children without free consent.

3. **Nazi Germany**

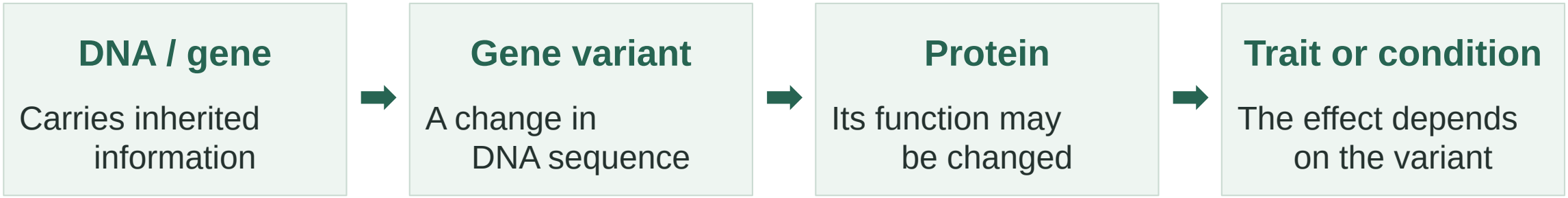
   Racial ideas and claims about inherited “defects” were used to support compulsory sterilization and persecution.

Lesson: Scientific language cannot justify discrimination or forced reproduction decisions.

b. Scientific basis: genes and molecules

How inherited differences can affect the body

DNA contains genes. Gene variants can change how proteins work and influence a trait or disease.



Example: Tay–Sachs disease

Some harmful variants in HEXA reduce the activity of the enzyme beta-hexosaminidase A. GM2 ganglioside then builds up in nerve cells and causes damage.

Many common conditions involve many genes plus environment, not one simple pathway.

b. Mechanism: a simple inheritance example

Two carriers of the same autosomal-recessive condition

Parent 1: A / a \ Parent 2: A / a	A	a
A	**AA** Without this variant	**Aa** Carrier
a	**Aa** Carrier	**aa** Affected genotype

For each conception

25%: affected genotype

50%: carrier

25%: without this variant

A = reference allele
a = pathogenic allele in this example

These are probabilities, not a fixed result for every four children.

b. Modern genetic technologies

Testing, selection, and editing are different

Carrier screening

Tests prospective parents for inherited risks.

PGT-M

Uses IVF to test embryos for a known single-gene condition.

Polygenic screening

Uses many variants to estimate complex disease risk.

Genome editing

Changes DNA; heritable changes may reach future generations.

A simplified PGT-M pathway

Genetic counselling ➡ **IVF** ➡ **Embryo testing** ➡ **Transfer decision**

Selection chooses among embryos; it does not repair their DNA.

c. Research paper 1

Karavani and colleagues • Cell

“Screening human embryos for polygenic traits has limited utility”

Aim

Can selecting embryos using polygenic scores reliably improve traits such as height?

Method

Statistical models were compared with genetic and family data.

Main finding

Predicted gains were limited and uncertain. The top-scoring embryo was not guaranteed the top outcome.

Limitations

Results depend on score accuracy, available embryos, and the trait. A score cannot predict a whole life.

Take-home point: A higher genetic score does not guarantee a “better” child.

c. Research paper 2

Lencz and colleagues • eLife

“Utility of polygenic embryo screening for disease depends on the selection strategy”

Aim and method

The study modelled disease risk and checked its predictions using virtual offspring based on real genetic data.

Two strategies

1. Exclude only very high-risk embryos.
2. Prioritize the embryo with the lowest score.

Main finding

Choosing the lowest-risk embryo could give larger relative risk reductions when enough viable embryos were available.

Limitations

It was not a long-term clinical trial. Results depend on the model, score accuracy, disease, and embryo number.

The paper studied complex diseases, including schizophrenia and Crohn’s disease.

c. What do the two papers tell us?

Compare the findings carefully

Question	Paper 1: traits	Paper 2: disease risk
Main focus	Predicted changes in traits such as height.	Predicted reduction in one complex disease risk.
Main message	Gains were limited and uncertain in the studied settings.	The strategy can greatly change the predicted benefit.
Shared limit	A prediction is not a guarantee for an individual child.	A model is not proof of long-term clinical benefit.

Example only: 10% risk → 7% risk

That is 3 percentage points lower, or a 30% relative reduction—not zero risk.

d. Benefits and limitations

Possible benefits depend on the practice and the evidence

Possible benefits

- Helps families understand inherited risks.
- May reduce the chance of a serious single-gene condition.
- Can give families more reproductive choices.
- Can support informed planning and counselling.

Limitations

- Many traits depend on genes and environment.
- Prediction does not guarantee health.
- IVF can be costly and physically demanding.
- Few embryos, test errors, and unequal access can limit benefit.

Strongest case: voluntary, reliable testing for a well-understood serious inherited condition.

e. Ethical implications

Scientific ability does not automatically make an action right

- Consent: People must be free to accept or refuse testing.
- Disability rights: A condition does not reduce a person’s worth.
- Fairness: Expensive tests may increase inequality.
- Privacy: Genetic information also affects relatives.
- Pressure: Families must not be blamed for declining selection.
- Embryos: People hold different views about their moral status.

Forced sterilization and reproductive discrimination violate basic rights.

Sources: [4], [5], [7]

e. When could selective reproduction be acceptable?

A simple way to think about the question

Reliable evidence

Does the test predict a clearly defined condition?

➡

Free choice

Can the family refuse without threats or penalties?

➡

Fair safeguards

Are benefits, burdens, rights, and alternatives explained?

Voluntary PGT-M for a serious inherited disorder	May be justified after counselling and careful assessment.
Forced sterilization or racial selection	Ethically unacceptable.
Promises to create a “perfect” child	Not supported by genetic prediction.

Sources: [4], [5], [7]

f. Future perspective

Better evidence and stronger safeguards are both needed

- Develop more accurate genetic tests for different populations.
- Study actual health outcomes, not only predicted scores.
- Explain risks in clear language, including absolute risk.
- Improve counselling, privacy protection, and fair access.
- Use strong oversight for heritable genome editing.
- Include people with genetic conditions in policy discussions.

Improve treatment and support as well as prevention.

g. Conclusion

A balanced answer to the topic

Scientifically

Using tests to guide embryo selection can reduce some inherited disease risks. Complex traits remain hard to predict. No test guarantees health.

Ethically

Voluntary choices may be acceptable with good evidence, counselling, and safeguards. Coercion and discrimination are not acceptable.

Selective reproduction may sometimes be justified for a serious inherited condition—but never as a way to rank human worth.

h. References

Two research papers and supporting sources

[1] Karavani, E., Zuk, O., Zeevi, D., et al. Screening human embryos for polygenic traits has limited utility. Cell, 179(6), 1424–1435.e8.

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